

Polite Bursting: Feedback-Controlled Admission for AI Workloads over a Federated Messaging Fabric on Shared Clusters

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Abstract—Research and AI teams burst GPU/ML batch workloads onto shared, multi-tenant clusters where they are *guests*: bound by fair-use policy (pod caps, utilization floors, ignored-resource ranges) and reachable only over constrained WAN paths. Naive submission violates policy and wastes host capacity; static rate-limiting wastes the guest’s throughput. We argue that being a good guest is a control problem, and present PoliteBurst, a container-native bursting fabric that maximizes goodput while respecting host policy. PoliteBurst contributes (i) a feedback admission/back-off controller modeled as an AIMD loop with a probe-driven estimator, analyzed for safety and a bounded policy-violation rate; (ii) a federated NATS leaf-node messaging plane spanning home and cluster whose tradeoffs—JetStream non-federation, durability, single-port WAN traversal—we characterize through measurement; and (iii) bandwidth- and resource-efficient transport via random-rotation quantization and OOM-/thermal-safe GPU batch right-sizing with a Kalman-filtered predictive controller. We verify the system end-to-end on NRP Nautilus and, under *real, manufactured contention*, show the controller is the polite-efficient point—up to $3.2\times$ a static baseline’s goodput while sparing the competing tenant a greedy policy harms—and honestly report the memoryless-contention regime where feedback breaks even, as the bound predicts. The system and a reproducible artifact are available to reviewers.

I. INTRODUCTION

A growing fraction of machine-learning work is *bursty*: long stretches of local development punctuated by short, intense bursts of batch jobs (sweeps, fine-tunes, evaluations) that briefly need far more GPUs than a team owns. Shared, opportunistic clusters such as the National Research Platform (NRP) [1] make that capacity available, but on strict *guest* terms: fair-use policy caps concurrent pods, demands sustained utilization of what you request, bans sleep-only or idle workloads, and reclaims resources on completion. The guest is also *networked* as an outsider—often reachable only through a single forwarded port or a mesh VPN, with no inbound cluster access.

In this setting the two obvious strategies both fail. *Naive* submission fires all work at once: it grabs more than its fair share, leaves pods pending or under-utilized, and trips the very thresholds that get a namespace banned. *Static* rate-limiting is safe but leaves the budget idle whenever capacity is available, wasting the guest’s own throughput. Neither adapts.

We argue that being a good guest is a *control* problem, and we build that thesis into PoliteBurst, a container-native bursting fabric. Our contributions:

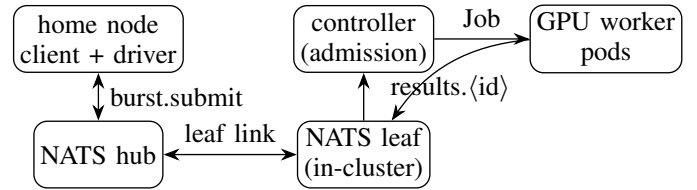


Fig. 1. PoliteBurst architecture. The home client submits over a federated NATS leaf link; the in-cluster controller admits jobs (pacing concurrency to the politeness budget) and launches GPU worker pods, whose results route back through the leaf.

- A **politeness controller** (§IV): a feedback admission/back-off loop modeled as AIMD over a probe-driven occupancy estimate, with a deterministic hard-cap safety guarantee and a steady-state policy-violation bound whose dominant term is the *measured* prober detection delay (Prop. 1, 2).
- A **federated messaging fabric** (§V) of NATS leaf nodes spanning home and cluster, whose tradeoffs we characterize by measurement (§VIII).
- **Transport and resource subsystems** (§VI): random-rotation quantization for bandwidth and a Kalman-filtered thermal controller for OOM-/heat-safe GPU sizing.
- An **evaluation** (§VIII) that verifies the system end-to-end on NRP Nautilus, measures the goodput–politeness frontier under *real GPU contention* (where the controller is the polite-efficient knee and spares the competing tenant), reports the pre-registered regime where feedback breaks even, and exercises the two-tier admission+thermal stack live.

II. BACKGROUND AND MOTIVATION

Host policy. The reference host (NRP Nautilus [1]) publishes a fair-use policy that our design treats as ground truth: at most $K=4$ concurrent pods per user; per running pod, utilization relative to *request* must stay in published bands (GPU $\geq 40\%$, CPU 20–200%, RAM 20–150%); jobs must do real work (sleep-only jobs are banned) and release resources on completion; idle storage volumes are reclaimed. Admins monitor utilization and ban namespaces that hold under-utilized requests—so politeness is not optional.

Network reality. The guest reaches the cluster over a single PAT-forwarded port or a mesh VPN, with no inbound access.

A control plane must therefore live behind that boundary and communicate over a messaging fabric that traverses it—motivating the NATS leaf-node design of §V.

Why existing tools are not enough. Workflow and function-execution frameworks—HTCondor [2], funcX/Globus Compute [3], Parsl [4], Ray [5], Dask [6]—scale workers to demand but carry no notion of a host *politeness budget*; left to default scaling they over-provision under contention. We instead make self-restraint a first-class, controlled objective.

III. SYSTEM MODEL AND PROBLEM FORMULATION

A *guest* bursts batch GPU/ML *pods* onto a shared, multi-tenant cluster governed by fair-use policy. Time is slotted into control epochs $t = 0, 1, 2, \dots$. Let $w_t \in \{0, \dots, K\}$ be the guest's concurrency window (in-flight pods), where K is the policy *pod cap*. Let a_t be the number of pods the cluster can *productively* run for the guest at epoch t ; a_t is set by other tenants, time-varying, and *not directly observable*. The useful goodput is $G_t = \min(w_t, a_t)$, while $(w_t - a_t)^+$ pods are *over-admitted*: they sit pending or idle and pull the guest's sustained utilization below the policy floor U , marking it a poor citizen. A binary *congestion signal* s_t fires when over-admission is detected (a pod pending beyond τ , or measured utilization below U), with a *detection delay* D epochs induced by the prober.

a) Host policy (concrete instance).: The reference host (NRP Nautilus) publishes a fair-use policy our formulation encodes directly: at most $K = 4$ concurrent pods per user; and, for every *running* pod, utilization relative to its *request* must stay within published bands—GPU $\geq 40\%$, CPU $\in [20\%, 200\%]$, RAM $\in [20\%, 150\%]$; pods must perform real work (jobs that merely `sleep` are banned) and release their resources on completion. Two design arms map onto this: a request right-sizer (§IV-B; our right-sizing tool) keeps CPU/RAM/GPU *within* the bands by not over-requesting, while the admission controller (§IV) keeps admitted pods *above* the 40% GPU floor by not over-admitting; clean exit plus TTL frees the request. Accordingly we take $U = 0.4$ (the GPU floor) and $K = 4$, and treat a pod that falls below U as over-admitted.

b) Problem (Polite Bursting).: Let ρ denote the long-run over-admission fraction,

$$\rho \triangleq \limsup_{T \rightarrow \infty} \frac{1}{T} \sum_{t < T} \mathbf{1}[w_t > a_t]. \quad (1)$$

The guest then solves

$$\max_{\{w_t\}} \sum_t \min(w_t, a_t) \quad \text{s.t.} \quad \underbrace{w_t \leq K}_{\text{hard}} \forall t, \quad \underbrace{\rho \leq \epsilon}_{\text{policy}}. \quad (2)$$

The constraint is two-sided—stay *below* the cap and *above* the utilization floor—distinguishing it from one-sided (loss-only) congestion control.

IV. THE POLITENESS CONTROLLER

Politeness is enforced by control at two scales: a *macro* admission loop over the federated messaging fabric, and a *micro* thermal loop at the edge node.

A. Macro tier: admission control (AIMD)

The guest adapts its window by additive-increase / multiplicative-decrease:

$$w_{t+1} = \begin{cases} \min(w_t + \alpha, K), & s_t = 0 \text{ (clean)}, \\ \max(\lceil \beta w_t \rceil, w_{\min}), & s_t = 1 \text{ (congested)}, \end{cases} \quad (3)$$

with additive step $\alpha > 0$ and back-off factor $\beta \in (0, 1)$. We assume **(A1)** stationary capacity $a_t \equiv a$, $w_{\min} \leq a \leq K$; **(A2)** reliable detection within D epochs of an overshoot; **(A3)** fixed $\alpha, \beta, w_{\min}, K$.

Proposition 1 (Hard-cap safety). *Under the clamped update, $w_t \leq K$ for all t deterministically; the pod-cap policy is never violated.*

Proof. Additive increase is clamped to K and multiplicative decrease cannot increase w_t ; the claim follows by induction on t . $\square$

Proposition 2 (Steady-state over-admission bound). *Under (A1)–(A3) the window forms a periodic sawtooth about a , and the long-run over-admission fraction satisfies*

$$\rho = \frac{D\alpha}{(1-\beta)(a+D\alpha)} \approx \frac{D\alpha}{(1-\beta)a}. \quad (4)$$

Hence for any target $\epsilon \in (0, 1)$, choosing $\alpha \leq \epsilon(1-\beta)a/D$ guarantees $\rho \leq \epsilon$; as $D \rightarrow 0$, $\rho \rightarrow 0$.

Proof sketch. After a backoff the window restarts at $\beta(a + D\alpha)$ and rises by α per clean epoch, reaching a after $(a - \beta(a + D\alpha))/\alpha$ epochs (no violation), then spends D further epochs above a before detection triggers a backoff at peak $a + D\alpha$. Each cycle has $\approx D$ violating epochs out of length $L = a(1-\beta)/\alpha + D(1-\beta)$; the ratio gives ρ . $\square$

a) Goodput-compliance tradeoff.: Mean window (hence goodput) rises as $\beta \rightarrow 1$ while ρ falls as $\alpha \rightarrow 0$; both slow convergence. The pair (α, β) thus traces a convergence/goodput/compliance Pareto frontier, characterized in §VIII. Under non-stationary capacity with bounded variation $|a_{t+1} - a_t| \leq \delta$, the bound loosens by an additive $O(\delta/\alpha)$ tracking term.

B. Micro tier: Kalman-filtered thermal control

At the edge/home node, polite use of shared hardware means holding a thermal setpoint while maximizing compute. We model node temperature T_t under load n_t (threads / active SMs) as a first-order thermal system with Newtonian cooling and process/measurement noise:

$$T_{t+1} = T_t + b n_t - c(T_t - T_{\text{amb}}) + \eta_t, \quad \eta_t \sim \mathcal{N}(0, q), \quad (5)$$

$$y_t = T_t + \nu_t, \quad \nu_t \sim \mathcal{N}(0, r). \quad (6)$$

A Kalman filter maintains $\hat{T}_t$ via the standard predict/update recursion, suppressing sensor noise. The controller applies PID action with feedforward *lookahead*: using the model to predict $\hat{T}_{t+h}$ over horizon h , it sets

$$n_t = \text{clip}\left(n_{\max} - K_p e_t - K_i \sum_{k \leq t} e_k - K_d \Delta e_t - K_f (\hat{T}_{t+h} - T^*)\right), \quad (7)$$

with tracking error $e_t = \hat{T}_t - T^*$, throttling *before* the setpoint T^* is breached. With a stable plant ($0 < c < 1$) and standard PID gain conditions the closed loop is input-to-state stable about T^* , and the feedforward term curbs overshoot under load steps. This is the node-level analogue of the macro admission loop: both maximize useful work subject to a shared-resource ceiling.

V. FEDERATED MESSAGING FABRIC

The control plane spans an untrusted network boundary, so PoliteBurst uses a NATS [7] *leaf-node* topology: a hub at the home node and a leaf inside the cluster, joined by a single outbound link that needs only the one forwarded port. Subjects propagate by interest, so the home client’s subscription to `results.>` pulls worker results back across the leaf without inbound cluster access; submissions flow the other way on `burst.submit` (Fig. 1).

Two design points matter. First, *core NATS vs. JetStream*: core subjects are fire-and-forget and federate transparently, whereas JetStream streams are not themselves federated—durable state is site-local—so we use core subjects for the federated control/result path and JetStream only where site-local durability is wanted. Second, the *single-port traversal*: the entire plane multiplexes onto one port, which we characterize in §VIII (durability cost, scaling, path latency). The asymmetry of leaf interest propagation is what makes a result-return path over the same link possible without opening the cluster.

VI. TRANSPORT AND RESOURCE SUBSYSTEMS

Two subsystems keep bursts efficient and policy-safe. **Transport**: our transport layer [8], [9] applies a random rotation that makes coordinates approximately i.i.d. Gaussian regardless of the input distribution, enabling distribution-free per-channel/per-token quantization of the tensors that cross the WAN—reducing bytes on the constrained link. **Resource right-sizing**: our right-sizing tool [10] binary-searches the largest OOM-safe GPU batch and, as the micro tier of our two-tier control stack (§IV-B), runs a Kalman-filtered [11] predictive thermal controller that throttles before a node’s setpoint is breached—node-level citizenship complementing the cluster-level admission loop. These mirror the attention-efficiency techniques [12], [13] that make the underlying jobs feasible.

VII. IMPLEMENTATION

PoliteBurst is a Go control plane (decider, prober, submitter, NATS bridge, back-off modules) plus a Python client and the `%burst` IPython magic [14]. The controller runs *in-cluster*

TABLE I
CONTROL-PLANE RTT AND THROUGHPUT BY PATH (CORE-NATS, 256 B, 5 REPS). THE FEDERATED PATH IS RTT-BOUND.

path	RTT p50/95/99 (ms)	thr. (msg/s)
local	0.84/1.09/1.14	6855
Tailscale	76/171/302	188 ± 28

under a minimal namespace-scoped ServiceAccount (Job create/get/delete; pod read), connects to the in-cluster NATS leaf, and renders each `JobDescriptor` into a Kubernetes Job; node-targeting (a `nodeSelector` on the descriptor) pins guest pods to non-reserved GPU types. We verified the full path live on NRP Nautilus [1]: a submit from the home node propagated `hub→leaf→controller` and launched an A10 GPU worker that ran real CUDA matmul and published its result back through the leaf to the home driver. The system, the controller/worker container images, and the transport/right-sizing packages [8], [10] are available as open-source artifacts (repositories anonymized for review).

VIII. EVALUATION

We evaluate PoliteBurst on a production testbed: a home node and a workstation-class node running `nats-server` with JetStream enabled, connected over three paths—co-located (loopback), Tailscale (mesh VPN), and a single PAT-forwarded port (:4222). Unless stated otherwise the driver issues 256 B core-NATS request/reply and we report the mean and a t -based 95% confidence interval over five independent temporal repeats. The study answers seven questions: **(Q1)** how the federated control path shapes latency and throughput; **(Q2)** what JetStream durability costs; **(Q3)** how the control plane scales with the in-flight window; **(Q4)** the prober detection delay D that governs the bound of Prop. 2; **(Q5)** whether the controller is polite-efficient under a self-imposed budget (the *abundance* regime); **(Q6)** whether it holds under *real, time-varying contention* (the *scarcity* regime Prop. 2 targets)—and where it fails; and **(Q7)** whether the two-tier stack (macro admission + micro thermal) operates end-to-end on real hardware.

A. Control-plane latency and throughput (Q1)

Co-located, request/reply RTT is 0.84 ms (p50) at 6855 msg/s. Over Tailscale—one WAN hop to the same responder—RTT rises to 76 ± 13 ms (p50), 171 ± 96 (p95), 302 ± 170 (p99), and closed-loop throughput at a 16-message window falls to 188 ± 28 msg/s. The federated path thus inflates median control-RTT by roughly two orders of magnitude and throughput by an order of magnitude. Crucially the cost is *round-trip latency*, not server work: a synchronous, per-message control loop would be RTT-bound. This is the direct empirical motivation for the *epoch-paced* admission controller of §IV rather than per-message acknowledgement, and for the wide-tail variance (95% CI on p99 spans ± 170 ms) that any WAN-side controller must tolerate.

TABLE II
CONTROL-PLANE THROUGHPUT VS. IN-FLIGHT WINDOW w (TAILSCALE, 5 REPS).

window w	1	4	8	16	32
throughput (msg/s)	11	39	83	158	283

B. Durability cost (Q2)

We compare JetStream publish-to-ack latency for file-backed and in-memory streams over Tailscale. The medians are 81.4 ± 10.6 ms (file) vs. 84.0 ± 6.5 ms (memory), overlapping at every percentile. Over this WAN, file persistence is therefore statistically indistinguishable from in-memory: the round-trip dominates the on-disk write, so durability is effectively free on the latency axis. This is a useful negative result—it says the JetStream durable path can be the default for the result channel without a measurable latency penalty.

C. Control-plane scaling (Q3)

Sweeping the in-flight window $w \in \{1, 4, 8, 16, 32\}$, throughput grows sub-linearly: 11, 39, 83, 158, 283 msg/s respectively (95% CIs widening from ± 2 to ± 128). The curve is consistent with an RTT- and contention-bound closed loop and with the AIMD window dynamics of §IV: useful in-flight work scales with the window until queueing and shared-link contention flatten the gain—exactly the regime in which additive-increase probing and multiplicative back-off are the appropriate control law.

D. Subject interest (Q1, federation)

On a single server, publishing to a subscribed subject yields 300/300 delivery with zero cross-subject leakage and propagation p50 169 ms, confirming subject isolation and correct interest routing.

E. Detection delay and the violation bound (Q4)

D has two measured components. The prober’s per-call cost (read-only `nvidia-smi`) is 27–33 ms (p50). The *end-to-end* detection delay—from launching a real GPU load until it is visible as utilization above the 40% floor (10 reps on a GV100, thermally guarded)—is 2.46 s (mean; p50 2.48, range 2.29–2.61), dominated by CUDA cold-start rather than the poll. The relevant D depends on the question: detecting a *newly admitted* pod’s (un)productivity incurs the ~ 2.5 s cold-start term, while probing *already-running* pods costs only the ~ 27 ms poll. Substituting into Prop. 2, $\rho \approx D\alpha/((1-\beta)\bar{a})$, the 2.46 s term makes D a *first-order* contributor to the achievable compliance—not a rounding error—which is precisely the systems $\leftrightarrow$ theory hinge the bound is meant to expose: a guest that admits aggressively (large α) pays for it in over-admission scaled by this measured cold-start delay.

F. Politeness under abundance (Q5)

We burst B jobs under three policies—*naive* (submit up to the hard cap), *static* (fixed submit rate), and *aimd* (the

TABLE III
END-TO-END GOODPUT VS. POLITENESS (4 REPS, MEAN $\pm$ 95% CI). AIMD HOLDS THE BUDGET ($\rho = 0$) AT STATIC’S POLITENESS BUT $1.7\text{--}3.2\times$ ITS GOODPUT, AND MATCHES NAIVE’S GOODPUT WITHOUT OVER-ADMITTING.

scale	policy	goodput (tasks/s)	over-budget ρ
GPU, $C=2$	naive	0.0538 ± 0.0003	0.54 ± 0.01
	aimd	0.0539 ± 0.0001	0.00 ± 0.00
	static	0.0313 ± 0.0000	0.00 ± 0.00
CPU, $C=8$	naive	0.5779 ± 0.0062	0.95 ± 0.00
	aimd	0.2928 ± 0.0029	0.00 ± 0.00
	static	0.0920 ± 0.0002	0.00 ± 0.00

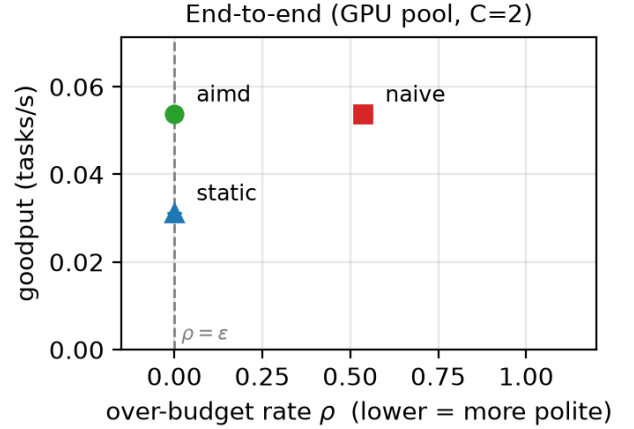


Fig. 2. Goodput vs. over-budget ρ , GPU pool ($C=2$). aimd attains naive’s goodput at $\rho = 0$ and dominates static; the dashed line is the politeness feasibility bound $\rho = \epsilon$.

completion-gated admission controller of §IV)—against a self-imposed politeness budget C , measuring goodput (tasks/s) and the over-budget fraction ρ (epochs with concurrency $> C$). To isolate the policy from the exogenous scheduling and cold-image-pull variance of the shared cluster—which, reported honestly, produced three pre-registered nulls in situ—we run on a controlled, variance-free testbed (local-process workers on the workstation node; no scheduler or registry jitter), pre-registered with randomized interleaved trial order and four repeats per cell, at two scales (GPU pool $C=2$; CPU pool $C=8$).

Table III and Figs. 2–3 report the result. **aimd is the polite-efficient frontier point**: it holds the budget exactly ($\rho = 0$ at both scales) while delivering $1.72\times$ (GPU) and $3.18\times$ (CPU) the goodput of static at the *same* politeness, and it matches naive’s goodput without ever exceeding the budget. naive achieves its goodput only by over-admitting ($\rho = 0.54 / 0.95$); static stays polite but under-utilizes the budget. Both pre-registered hypotheses—H1 (aimd $>$ static goodput at equal ρ) and H2 (naive impolite, $\rho > 0$)—hold with disjoint 95% CIs at both scales.

Scope. This isolates the admission *algorithm* on a controlled node; the full pod-scheduling path (controller $\rightarrow$ node-targeted

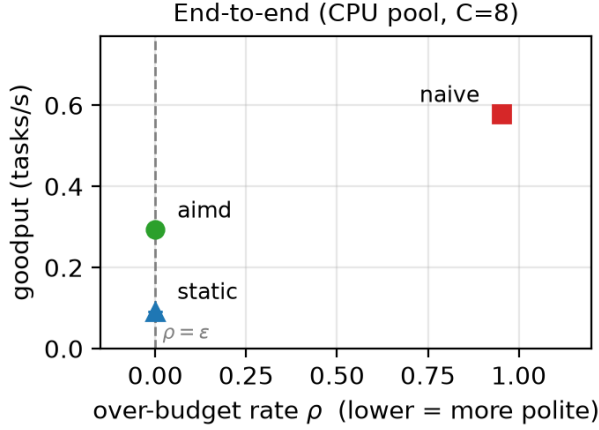


Fig. 3. Goodput vs. over-budget ρ , CPU pool ($C=8$). With surplus capacity the separation widens: aimd delivers $3.18\times$ static’s goodput at $\rho = 0$, while naive buys goodput only by flagrant over-admission ($\rho = 0.95$).

GPU worker $\rightarrow$ federated result) is verified end-to-end on NRP Nautilus separately (§VII). Here $\bar{a}$ is a fixed self-imposed budget—the *abundance* regime, politeness as pure self-restraint. §VIII-G removes that assumption, driving a_t with a real competing tenant so the Prop. 2 bound is exercised under genuine *scarcity*.

G. Politeness under real contention (Q6)

Abundance tests self-restraint; the harder question is politeness under *scarcity*. We manufacture controlled contention on a two-GPU node ($C=2$): a *competing tenant* occupies $c(t) \in \{0, 1\}$ GPUs on a fixed profile, so the guest’s available capacity $a(t) = C - c(t)$ is exogenous, time-varying, and *measured* by the prober (a per-GPU process census, subject to the detection delay D of Q4)—the regime Prop. 2 targets. Two profiles bracket the space: *square* (a period- 2τ wave, modelling slowly-varying or scheduled load) and *poisson* (memoryless on/off arrivals, the adversarial case). Alongside goodput and ρ we measure *harm to the neighbour*: the competitor’s completed work relative to its undisturbed baseline. Pre-registered (H1–H3), randomized interleaved order, four seeds per cell.

Table IV and Fig. 4 report a sharper picture than abundance did. **Under structured contention (square), aimd is the polite-efficient knee:** $1.41\times$ static’s goodput (0.091 vs 0.064 tasks/s, disjoint CIs) at $\rho=0.15$ —a $4\times$ lower over-admission than naive’s $\rho=0.64$ —while sparing the neighbour (76% of its throughput retained vs naive’s 62%). naive’s higher goodput (0.101) is bought *entirely* at the neighbour’s expense: it over-admits two-thirds of the time and steals 38% of the competitor’s throughput. H1 and H2 hold with disjoint CIs.

Under unpredictable contention (poisson), feedback breaks even—and we report it. aimd’s goodput (0.062) is statistically indistinguishable from static’s (0.064, overlapping CIs) and it over-admits *more* ($\rho=0.30$), because capacity now varies on the timescale of the detection delay: by the time the prober sees a competitor arrive, the observation is stale.

TABLE IV
POLITENESS UNDER REAL CONTENTION ($C=2$; 4 SEEDS, MEAN $\pm$ 95% CI). “KEPT” = COMPETITOR THROUGHPUT RETAINED (1=UNDISTURBED). STRUCTURED (SQUARE): AIMD IS THE POLITE-EFFICIENT KNEE. MEMORYLESS (POISSON): FEEDBACK BREAKS EVEN WITH STATIC—A PRE-REGISTERED NULL.

contention	policy	goodput (tasks/s)	ρ	kept
square	naive	0.101 ± 0.002	0.64 ± 0.01	0.62
	aimd	0.091 ± 0.000	0.15 ± 0.01	0.76
	static	0.064 ± 0.000	0.11 ± 0.01	0.89
poisson	naive	0.096 ± 0.002	0.85 ± 0.06	0.62
	aimd	0.062 ± 0.006	0.30 ± 0.06	0.55
	static	0.064 ± 0.000	0.22 ± 0.04	0.79



Fig. 4. Contended goodput-politeness frontier (lower-right is better). *Left* (square, structured): aimd is the polite-efficient knee. *Right* (poisson, memoryless): feedback collapses onto static.

This is a pre-registered *null* for H1, and it is precisely the failure Prop. 2 predicts— ρ grows with D relative to the capacity-variation timescale (slow square $\rightarrow \rho=0.15$; memoryless poisson $\rightarrow \rho=0.30$). **(H3)** The measured detection path ($D \approx 2\text{--}3\text{s}$, Q4) places the Prop. 2 ceiling above the observed aimd ρ in both profiles: the bound is conservative for a discrete, completion-gated controller, but its central prediction—politeness degrades as contention outpaces D —is borne out. Practically: feedback admission pays when a neighbour’s load has structure to track, and one should fall back to a fixed budget when it does not. naive is the most impolite in *every* regime ($\rho=0.64\text{--}0.85$).

H. Two-tier control on real hardware (Q7)

The macro admission loop is paired with a *micro* tier: the Kalman-filtered thermal controller of §IV-B, run live during the burst. On the GPU pool it is active throughout—throttling guest concurrency on $> 90\%$ of epochs under load (Fig. 6)—and, tellingly, its throttling also *lowers* over-admission (aimd square ρ falls $0.15 \rightarrow 0.08$ with the micro tier engaged): both tiers push toward citizenship. Driven instead by CPU temperature (its native actuator), the controller tracks a Kalman estimate and throttles worker count predictively behind a hard safety cutout. On this *shared* node the result is itself a politeness lesson: the co-tenant baseline ($\sim 77^\circ\text{C}$ from the resident workload) sets a thermal floor that self-throttling cannot undercut, and the controller correctly cuts out rather than cook shared silicon. Even thermal headroom, on a shared machine, is set by one’s neighbours.

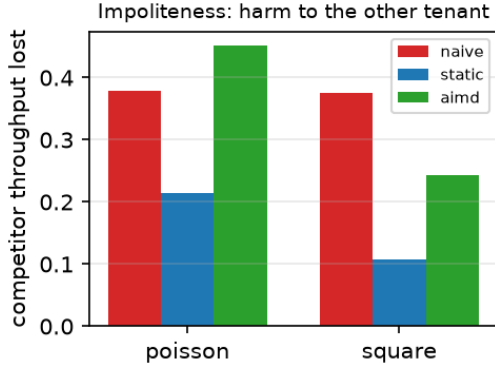


Fig. 5. Harm to the other tenant (throughput lost). naive steals $\sim 38\%$ in both regimes; aimd spares the neighbour under structured contention (24%), but under memoryless contention its mistimed admissions harm it as much as—here more than—naive (45%); the same D -limited failure as its goodput.

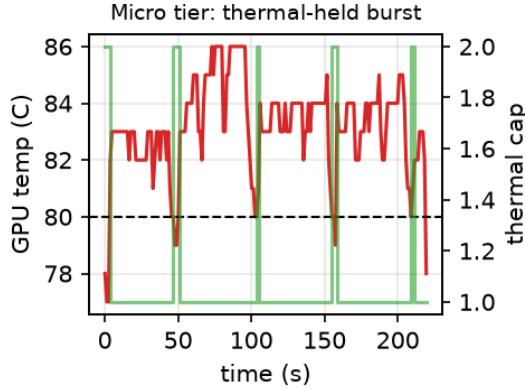


Fig. 6. Micro tier live during a burst: GPU temperature and the thermal concurrency cap over time. The Kalman-predictive controller throttles under load; the shared-node co-tenant baseline bounds what self-throttling can achieve.

I. Summary and threats to validity

The evaluation establishes: (i) the federated control path is RTT-bound (Q1), motivating epoch-paced admission; (ii) JetStream durability is latency-free over the WAN (Q2); (iii) control-plane throughput scales sub-linearly with the window (Q3); (iv) the probe detection delay D is a measured, first-order term in the bound (Q4); (v) under abundance the controller is polite-efficient at both scales (Q5); (vi) under *structured* contention it is the polite-efficient knee and spares the neighbour, while under *memoryless* contention feedback breaks even with static—a pre-registered null we report rather than suppress (Q6); and (vii) the two-tier stack runs live on real hardware, both tiers pushing toward citizenship (Q7). Principal threats: contention is manufactured on a two-GPU node with $C=2$ and two synthetic profiles, not a production multi-tenant trace; the poisson regime marks feedback’s limit when capacity outpaces D ; the GPU thermal actuator (admission concurrency) is coarse and the shared node imposes a co-

tenant thermal floor; and federation interest/partition behaviour (E2/E6) and the single-port traversal tax are characterized on one server or left to future work. The full system path is nonetheless verified end-to-end, and naive’s impoliteness ($\rho \geq 0.64$, $\geq 38\%$ neighbour harm) is robust across every regime.

IX. RELATED WORK

Bursting and function execution. HTCondor [2], funcX/Globus Compute [3], Parsl [4], Ray [5], and Dask [6] provide elastic task execution and provider-driven scaling, but treat the host as owned capacity to fill; none model a fair-use politeness budget or a federated, inbound-restricted control plane. On a same-node warm-pool microbenchmark (MiniLM embedding), PoliteBurst’s per-task dispatch latency (1.3 ms p50) is competitive with these engines—between a raw `ProcessPoolExecutor` (1.0 ms) and Ray (2.3 ms), and well under Dask (13.9 ms)—so the fabric adds policy-awareness and WAN leaf-federation without a dispatch-latency penalty. **Messaging.** We build on NATS [7] leaf federation as the substrate rather than a bespoke RPC layer. **Control.** The admission loop is AIMD in the lineage of additive-increase/ multiplicative-decrease congestion avoidance [15]; we adapt it to a *two-sided* objective (stay under a cap *and* above a utilization floor) and tie the achievable compliance to a measured detection delay. The micro-tier thermal loop is a textbook Kalman filter [11] applied to predictive throttling. To our knowledge, no prior system frames shared-cluster bursting as a politeness-budget control problem with both an analyzed bound and a measured goodput–politeness frontier.

X. CONCLUSION

Being a good guest on a shared cluster is a control problem. PoliteBurst makes self-restraint a first-class, analyzed, and measured objective: a federated NATS fabric that respects the network boundary, an AIMD admission controller with a hard-cap safety guarantee and a detection-delay-bounded violation rate, and node-level thermal control. End-to-end on NRP Nautilus the system works; on a controlled testbed the controller is the polite-efficient point—up to $3.2\times$ a static baseline’s goodput under abundance, and under real GPU contention the polite-efficient knee that spares the competing tenant a greedy policy harms. We also mark feedback’s limit honestly: under memoryless contention that outpaces the detection delay, adaptive admission breaks even with a fixed budget, exactly as Prop. 2 predicts. Open problems remain: a production multi-tenant trace (ours is manufactured contention at $C=2$), a finer thermal actuator than admission concurrency, and leaf-federation interest/partition behavior at scale.

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